CSAI 202

Introduction to Database Systems



Lec 02

Database System Concepts and Architecture



Outline

- Data Models
- Schema, Instances, and States
- Three-Schema Architecture
- DBMS Languages and Interfaces
- Multi-User DBMS Architecture

Data Models

- Data Model:
  - ★*A set of concepts* to describe the *structure* of a database, the *operations* for manipulating these structures, and certain *constraints* that the database should obey.

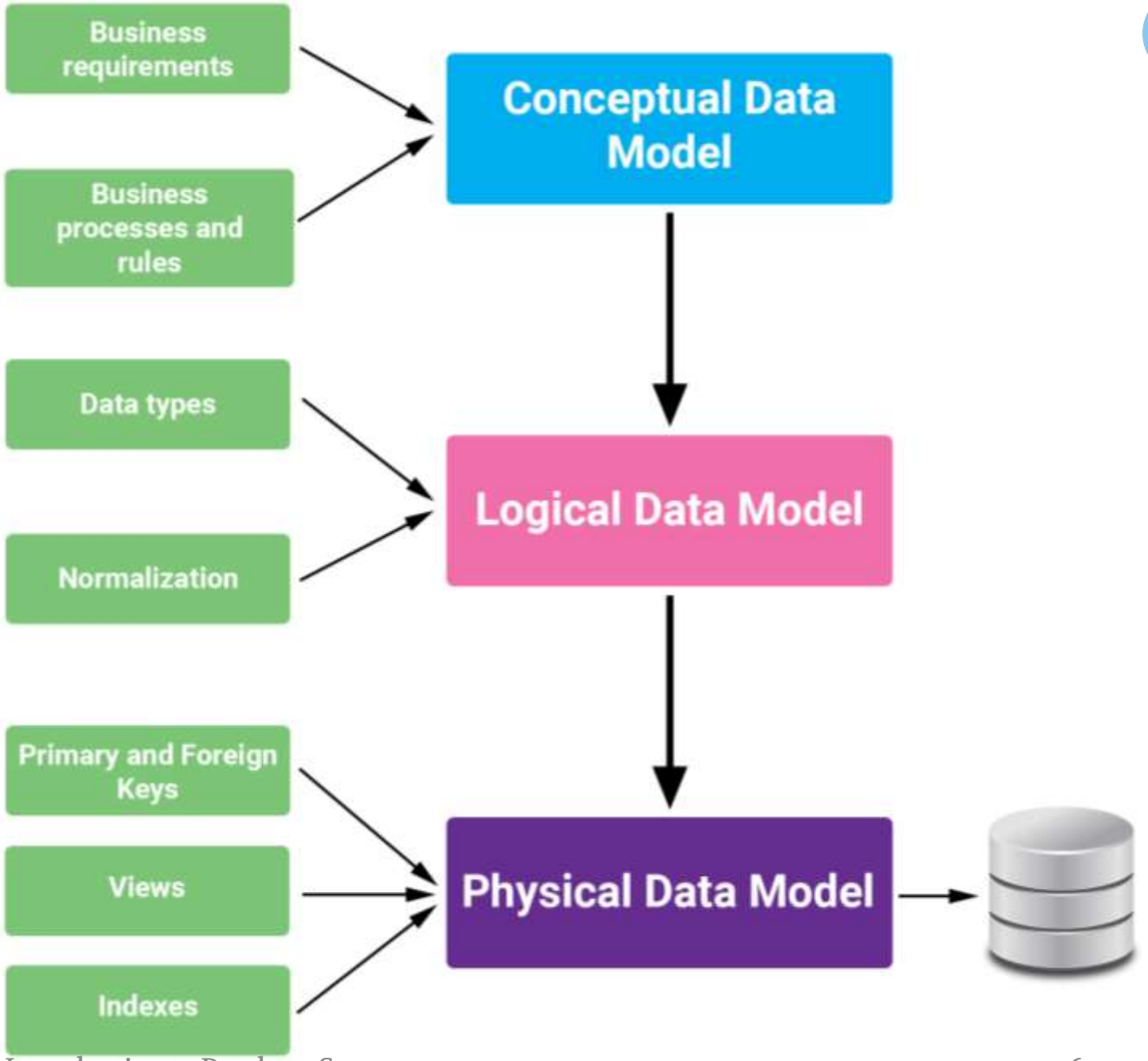
Structure and Constraints

- Data Model Structure:
  - Constructs are used to define the database structure
  - Constructs typically include
    - *elements* (and their *data types*) as well as groups of elements (e.g. *entity, record, table*)
    - *relationships* among such groups
- Data Model Constraints:
  - *Constraints specify some restrictions on valid data*; **these constraints must be enforced at all times**

Operations

- These operations are used for specifying database retrievals and updates by referring to the constructs of the data model.
- Operations on the data model may include
  - ★ ***basic model operations***
    - (e.g. generic insert, delete, update)
  - ★ ***user-defined operations***
    - (e.g. *compute_ student_ gpa, update_ inventory*)

Categories of Data Models



Categories of Data Models

- Conceptual (high-level, semantic) data models:
  - Provide concepts that are close to the way many users perceive data.
  - (Also called ***entity-based*** or ***object-based*** data models.)
  - ***Example Entity – Relationship Model (ER diagram)***

Categories of Data Models

- Implementation (representational) data models:
  - Provide concepts that fall between the above two, used by many commercial DBMS implementations (e.g. relational data models used in many commercial systems).
  - **Hide many details of data storage on disk BUT can be implemented directly on a computer system.**
  - *Example: Relational Data model*

Categories of Data Models

- Physical (low-level, internal) data models:
  - Provide concepts that describe details of *how data is stored in the computer*.
  - These are usually specified through DBMS design and administration manuals

Categories of Data Models

- Self-Describing Data Models
  - ★ Combine the description of data with the data values.
    - *Examples include XML and some NoSQL system*

Schemas vs. Instances

- Database Schema:
  - The *description* of a database.
  - Includes descriptions of the database *structure, data types*, and the *constraints* on the database.
- Schema Diagram:
  - An *illustrative* display of (most aspects of) a database schema.
- Schema Construct:
  - A *component* of the schema or an object within the schema
  - **e.g., STUDENT, COURSE.**

Schemas vs. Instances

- Database State:
  - The *actual data stored* in a database at a *particular moment in time*.
  - This includes the collection of all the data in the database.
  - Also called **database instance** (or occurrence or snapshot).
    - The term instance is also applied to individual database components
  - *e.g. record instance, table instance, entity instance*

Schemas vs. Instances

- Database State:
  - Refers to the content of a database at a moment in time.
- Initial Database State:
  - Refers to the database state when it is initially loaded into the system.
- Distinction
  - The *database schema* changes very infrequently.
  - The *database state* changes every time the database is updated.
  - *Schema* is also called *intension*.
  - *State* is also called *extension*.

Example of a Database Schema

STUDENT

Name	Student_number	Class	Major

COURSE

Course_name	Course_number	Credit_hours	Department

PREREQUISITE

Course_number	Prerequisite_number

SECTION

Section_identifier	Course_number	Semester	Year	Instructor

GRADE_REPORT

Student_number	Section_identifier	Grade

Figure 2.1
Schema diagram for the database in Figure 1.2.

Example of a database state

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

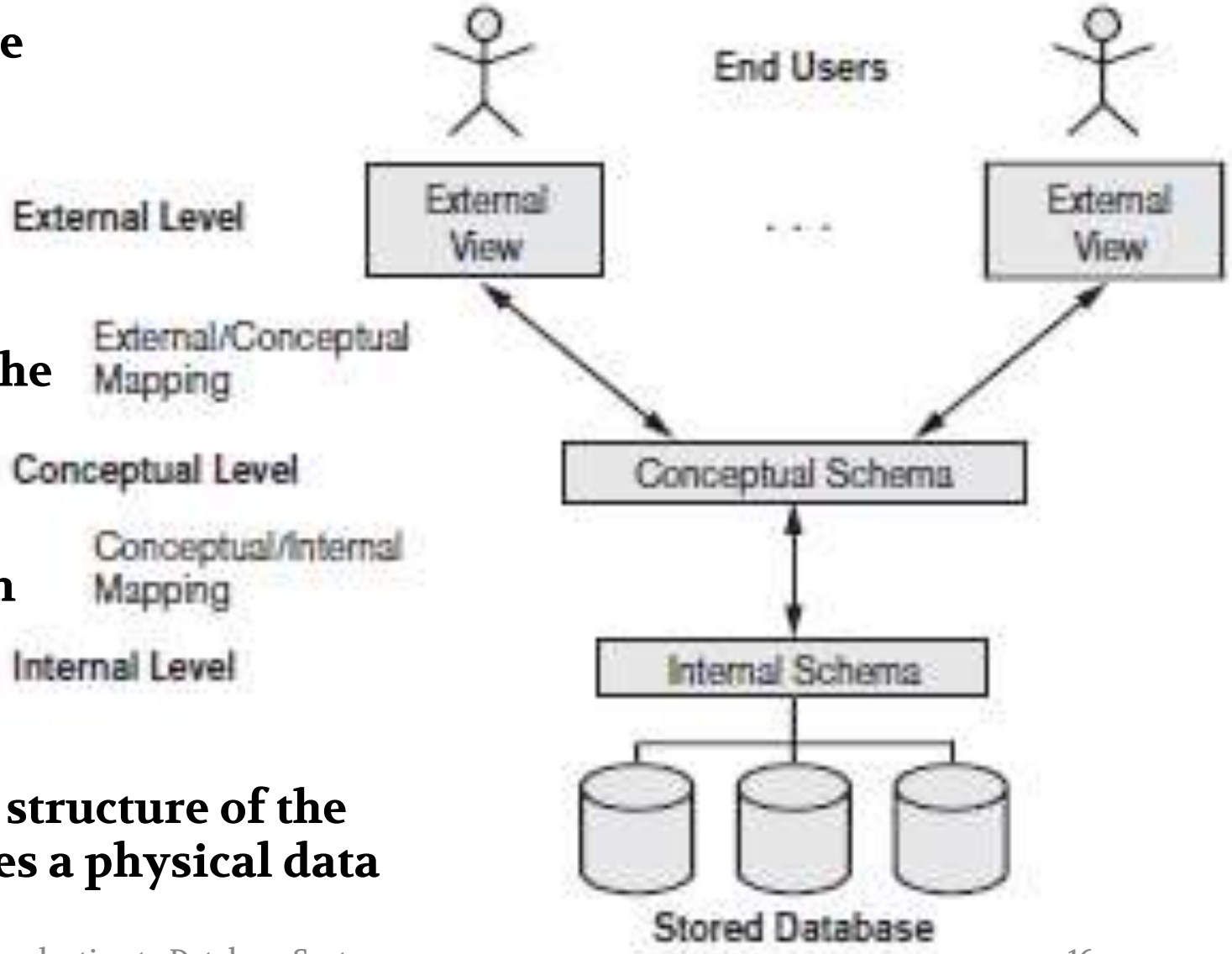
Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

The Three-Schema Architecture

External: at the external level to describe the various user views.
uses the same data model as the conceptual schema

Conceptual : Describes the structure of the whole database
Concentrate on describing the entities , data types, relationships,…
Uses a conceptual or an implementation data model.

Internal: Describes the physical storage structure of the database and access paths (Typically uses a physical data model)



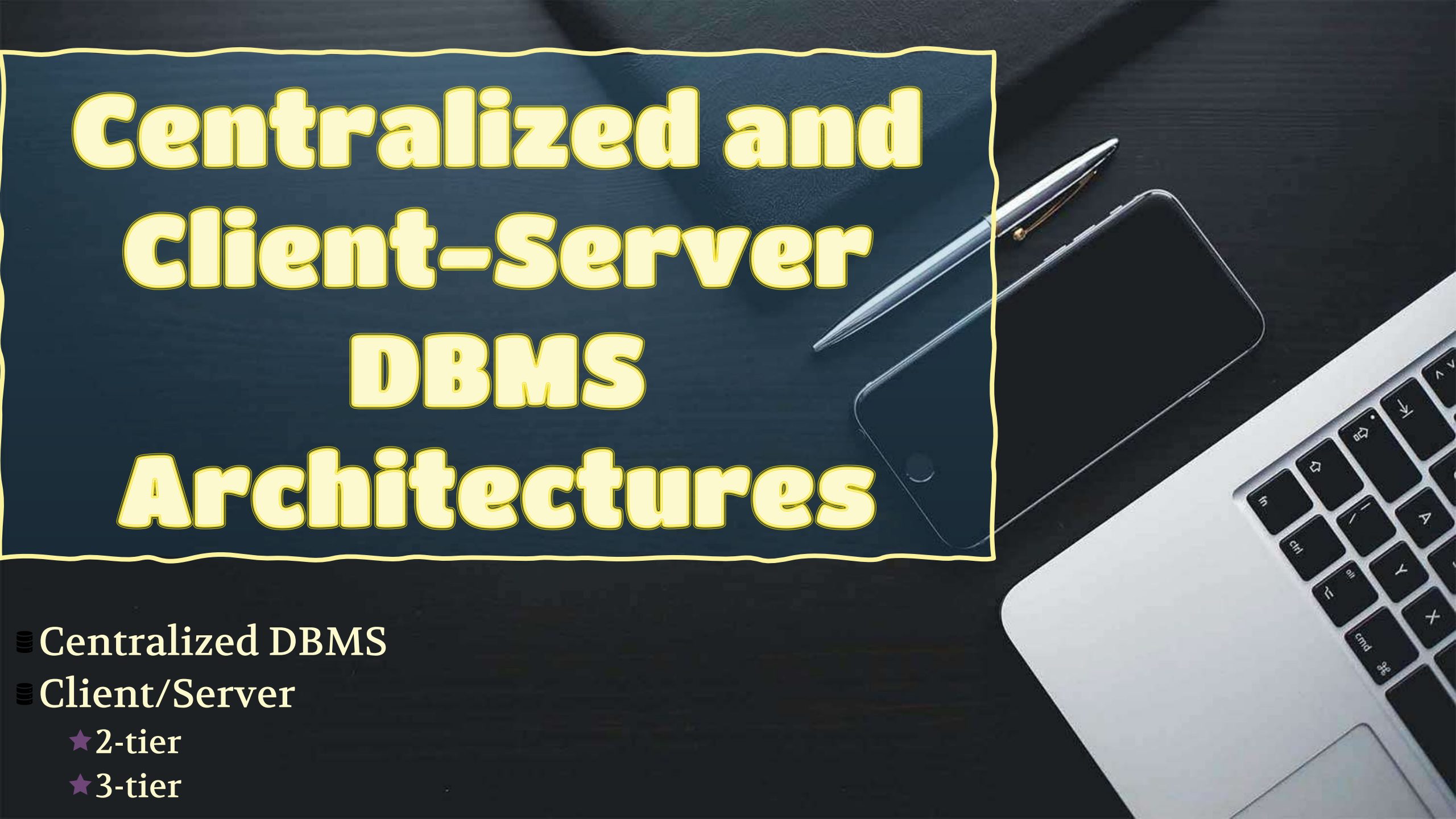
DBMS Languages

- Data Definition Language (DDL):
  - Used by the DBA and database designers to specify the conceptual schema of a database.
  - In many DBMSs, the DDL is also used to define internal and external schemas (views).
- Storage Definition Language(SDL):
  - specify the internal schema

DBMS Languages

- View Definition Language (VDL):
  - specify the external schema
- Data Manipulation Language (DML):
  - Used to specify database retrievals and updates
  - DML commands (data sublanguage) can be ***embedded*** in a general-purpose programming language (host language), such as COBOL, C, C++, or Java.
    - A library of functions can also be provided to access the DBMS from a programming language
  - Alternatively, stand-alone DML commands can be applied directly (called a ***query language***).

Centralized and Client-Server DBMS Architectures

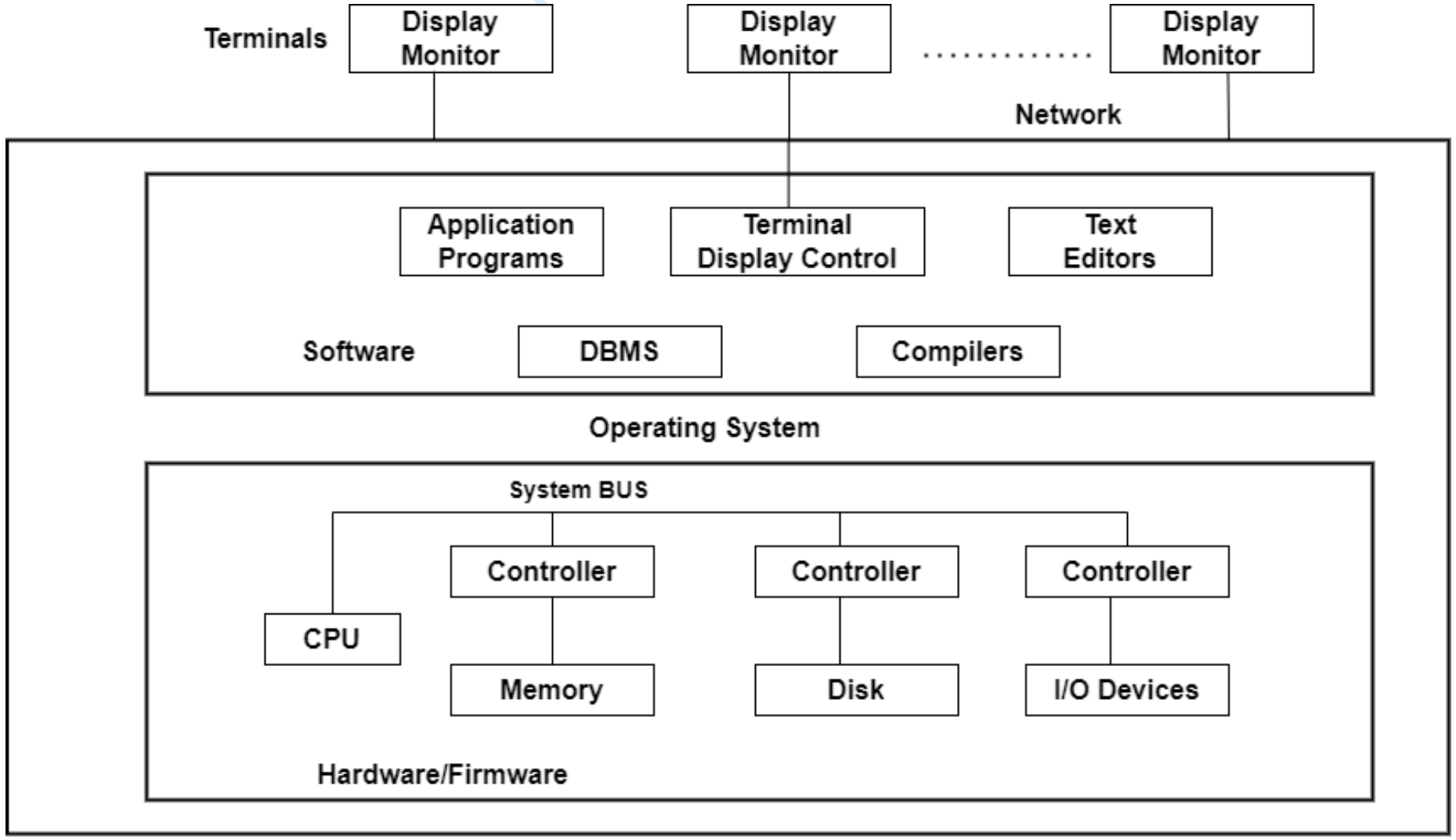


- Centralized DBMS
- Client/Server
  - 2-tier
  - 3-tier

Centralized DBMS Architectures

- Combines everything into single system including- DBMS software, hardware, application programs, and user interface processing software.
- User can still connect through a remote terminal – however, all processing is done at centralized site.

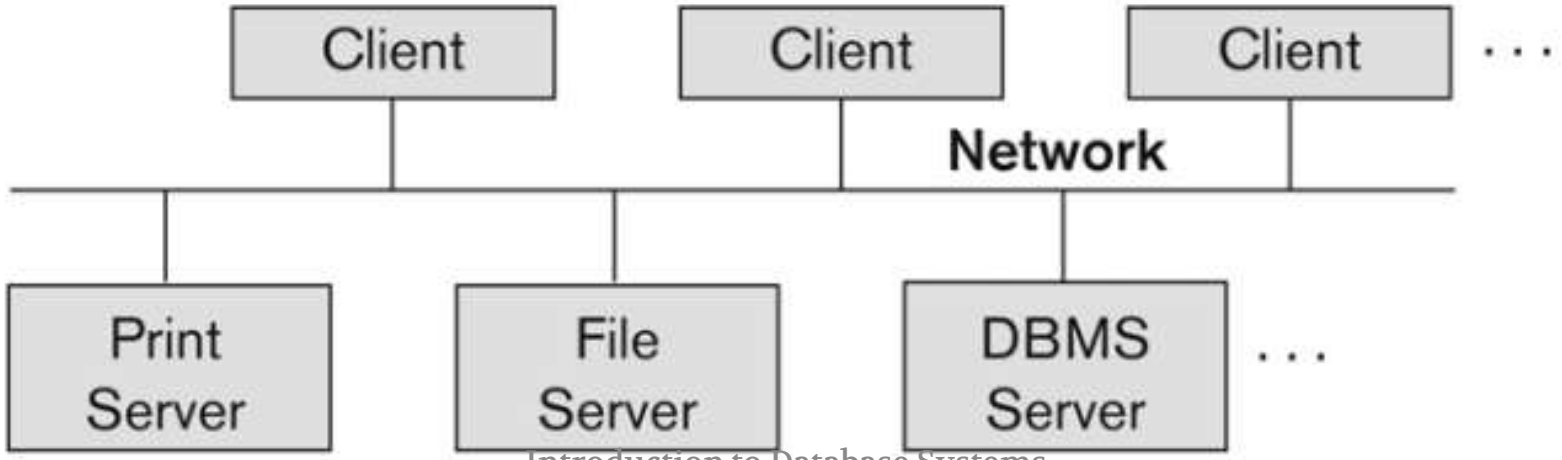
Centralized DBMS Architectures



Centralized architecture of DBMS

Basic 2-tier Client-Server Architectures

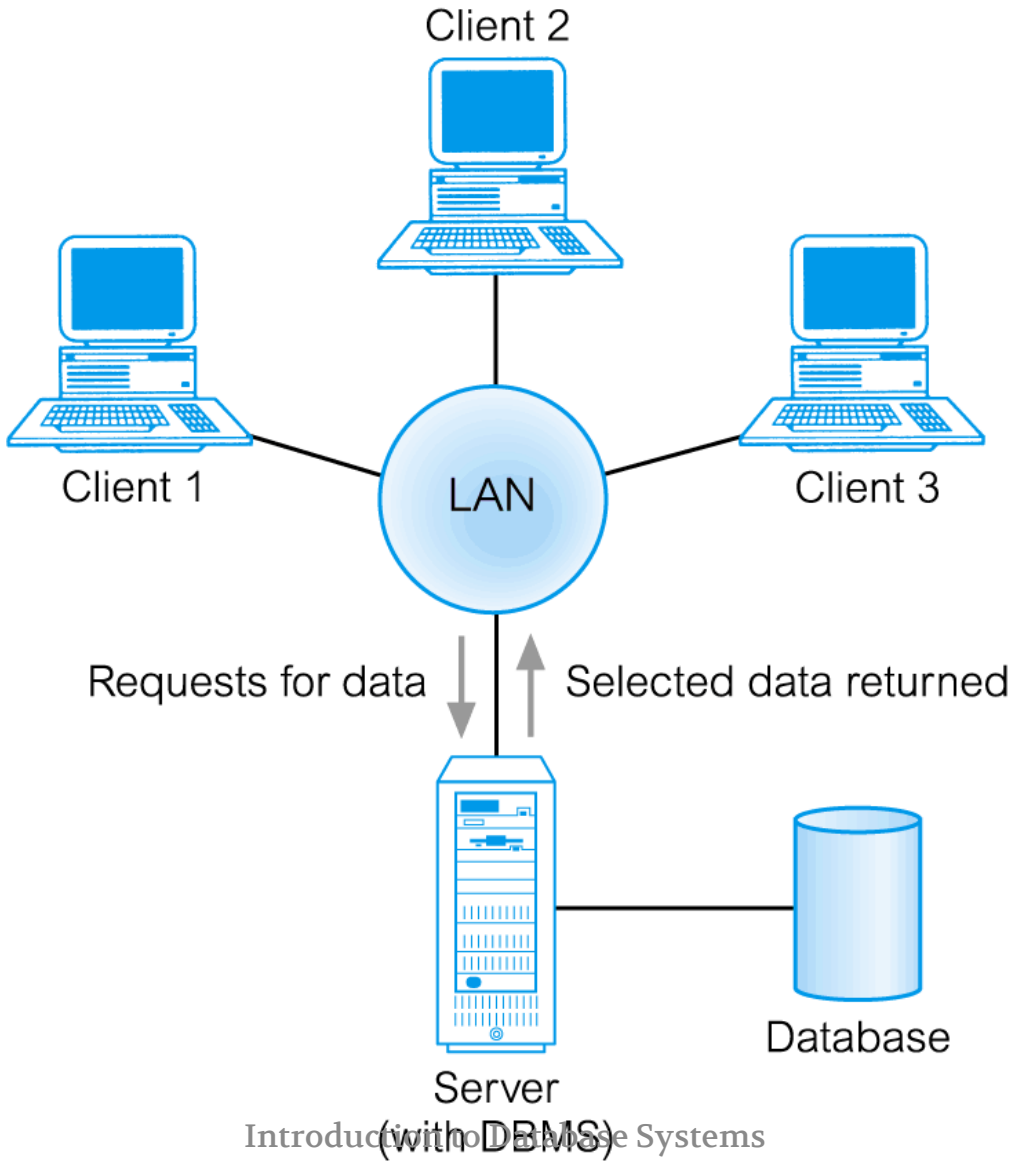
- Specialized Servers with Specialized functions
  - Print server
  - File server
  - DBMS server
  - Web server
  - Email server
- Clients can access the specialized servers as needed



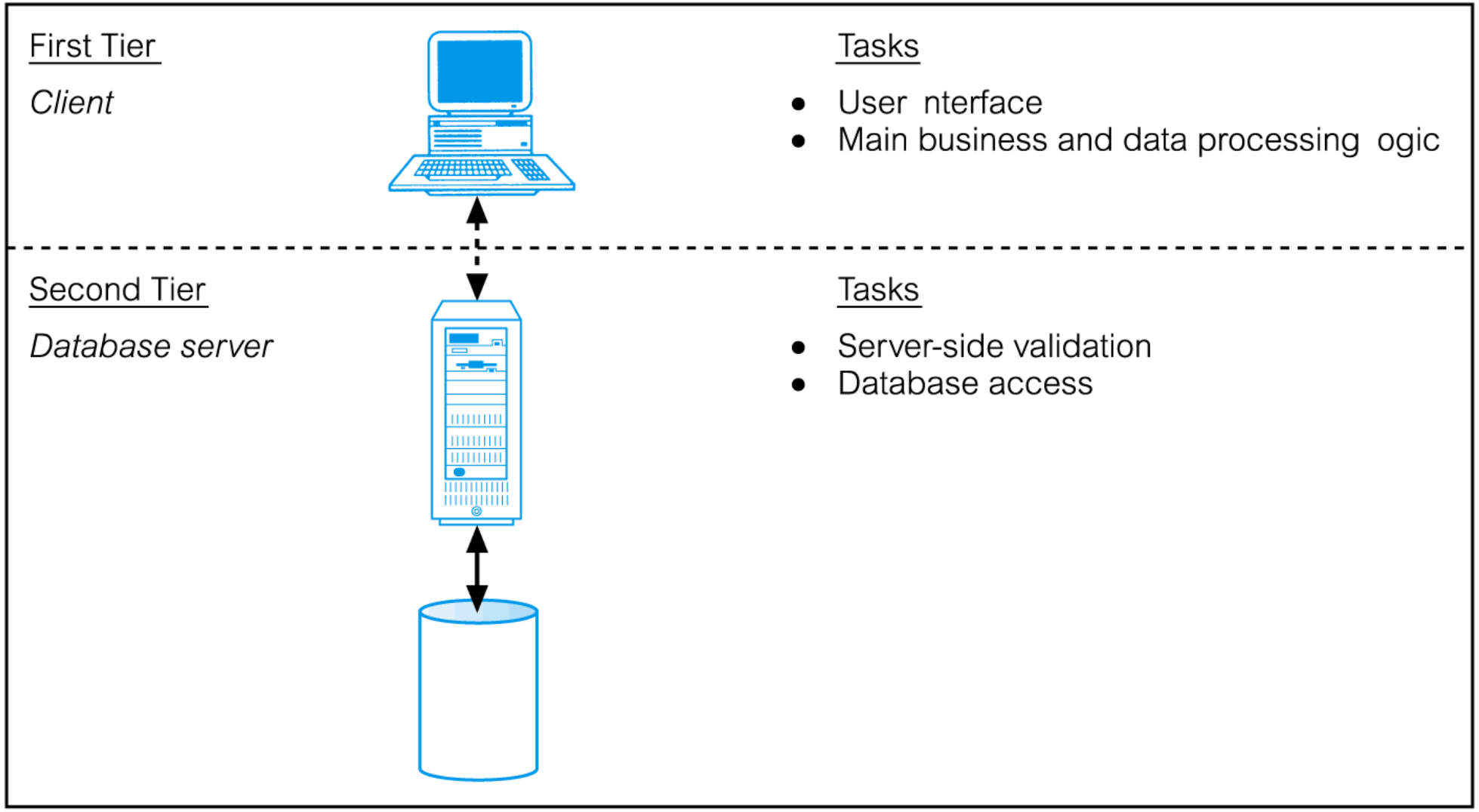
Basic 2-tier Client-Server Architectures

- Client
  - Provide appropriate interfaces through a client software module to access and utilize the various server resources.
  - Connected to the servers via some form of a network.
- Server
  - Provides database query and transaction services to the clients
  - Relational DBMS servers are often called SQL servers, query servers, or transaction servers
  - Applications running on clients utilize an Application Program Interface (**API**) to access server databases via standard interface
  - Client and server must install appropriate client module and server module software.

Traditional Two-Tier Client-Server



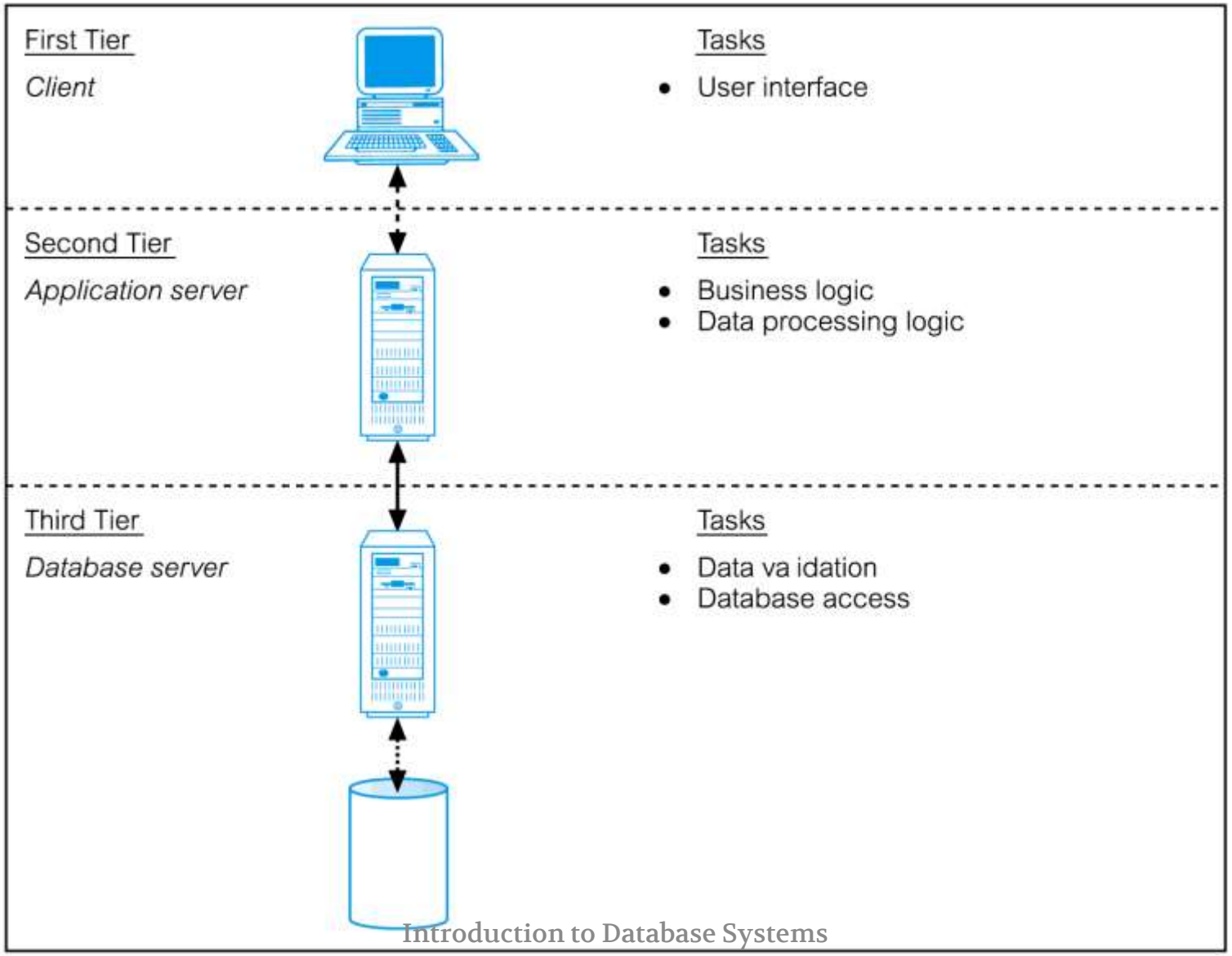
Traditional Two-Tier Client-Server



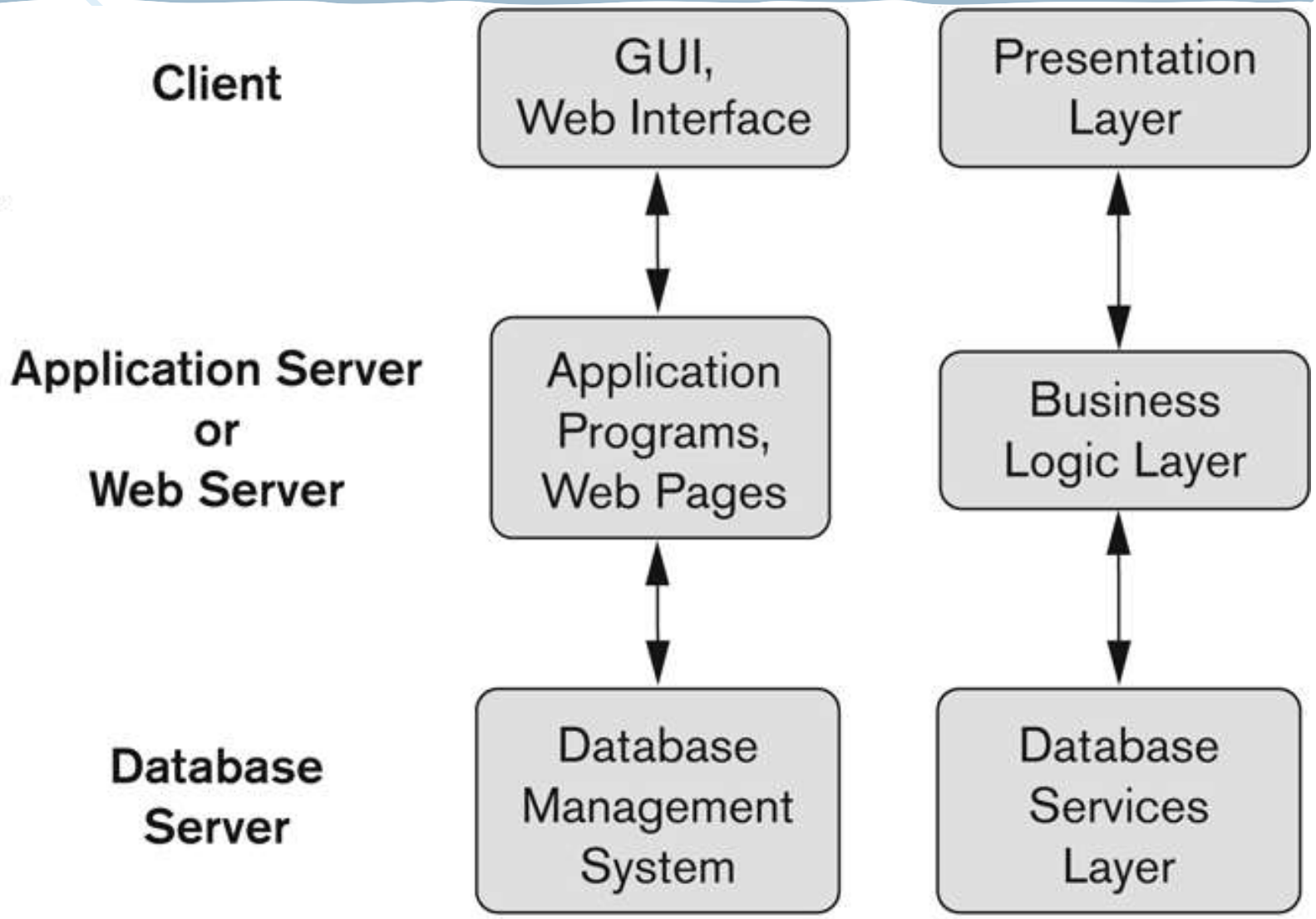
Three Tier Client–Server Architecture

- Common for *Web applications*
- Intermediate Layer called **Application Server** or **Web Server**:
  - Stores the web connectivity software and the business logic part of the application used to access the corresponding data from the database server
  - Acts like a conduit for sending partially processed data between the database server and the client.
- Three-tier Architecture Can Enhance Security:
  - Database server only accessible via middle tier
  - Clients cannot directly access database server
  - Clients contain user interfaces and Web browsers
  - The client is typically a PC or a mobile device connected to the Web

Three-Tier Client-Server



Three-tier client-server architecture



Summary

- Data Models
- Schema, Instances, and States
- Three-Schema Architecture
- DBMS Languages and Interfaces
- Multi-User DBMS Architecture

Thank you

Questions?

